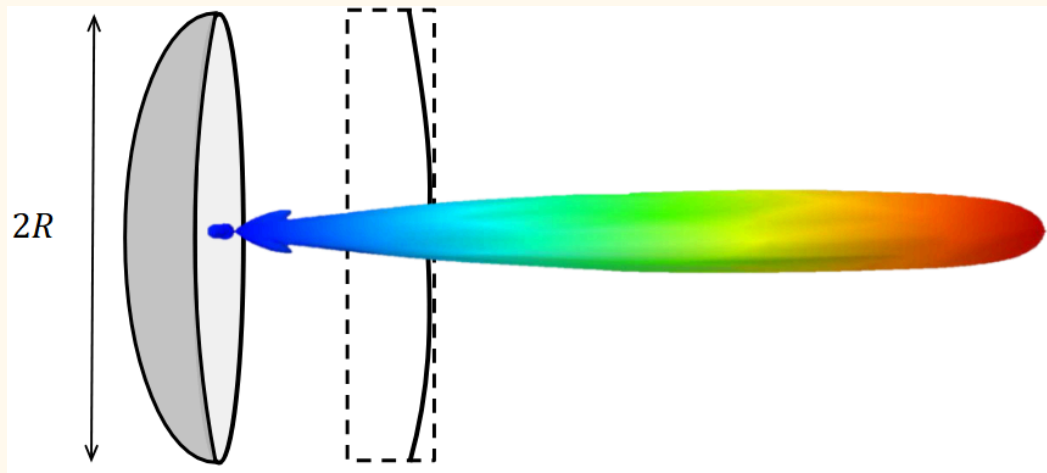


Parabolic Reflector Antennas (Dish)



Electrically large reflector (biggest dimension $> 10\lambda$)

$$HPBW = \frac{a}{2R/\lambda} \quad a \text{ is a constant: } 57.3 \text{ to } 70$$

$$G_{realised, max} = e_{ant} e_{ap} D_{max} = e_{ant} e_{ap} \frac{4\pi A_{physical}}{\lambda^2} = e_{ant} e_{ap} \frac{4\pi^2 R^2}{\lambda^2}$$

$e_{ap} \sim 0.55 \text{ to } 0.70$ **NOT THE SAME AS a**

